Erasure-Aware Byzantine-Robust Aggregation for Hierarchical UAV-Swarm Federated Learning: When Link Telemetry Hurts

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Abstract—Federated learning (FL) in UAV swarms must tolerate compromised clients and incomplete updates. We study hierarchical FL with packet erasures driven by a measured air-to-ground trace, assumed reference signal-to-noise ratio (SNR) anchors and virtual neighboring base stations. A trusted-reference reputation aggregator is tested with three levels of channel awareness: none, packet-erasure masking, and link telemetry inside the reputation update. A prespecified paired study comprises 720 runs (six aggregators, a clean setting and three model-poisoning attacks, three channel conditions, 10 seeds). The primary telemetry stress interaction is -2.7 percentage points (pp; 95% paired-seed bootstrap confidence interval (CI) $[-3.9, -1.5]$). Under attack at the 5 dB anchor, masking improves accuracy by 4.2 pp ($[2.2, 5.9]$) over no-context aggregation, whereas adding telemetry changes it by -3.5 pp ($[-5.1, -2.1]$) relative to mask-only, accompanied by a rise in honest-update rejection from 13.8% to 41.2%. A constant-evidence drift analysis suggests that asymmetric trust recovery lowers reputation on weak links. A separate 240-run follow-up finds that fresh-only averaging exceeds reuse of old parameters by 5.39 pp on clean stressed links ($[4.11, 6.74]$); neither rule tolerates the tested sign-flip attack.

Index Terms—Federated learning, UAV swarm, Byzantine robustness, packet erasure, hierarchical aggregation, reputation.

I. Introduction

UAV swarms increasingly train perception models collaboratively, and FL [1] keeps each UAV's imagery on board while exchanging only model updates; it has been applied to on-board UAV-swarm missions [2]. Two impairments act on those updates at once. First, air-to-ground links fade, hand over between base stations (BSs) and miss deadlines, so updates are erased or arrive incomplete; for erased updates, the server can average only fresh ones or reuse stale ones [3]. Second, captured or spoofed UAVs can send poisoned updates. Robust rules such as Krum [4], coordinate-wise median [5] and FLTrust [6] address the second problem, in their original form for complete update vectors. Byzantine-robust designs also exist for analog over-the-air aggregation, including UAV-assisted architectures [7], [8], and anti-jamming designs treat malicious clients as jammers [9]. Reputation-based defenses suit persistent swarm members, yet they can themselves be gamed [10].

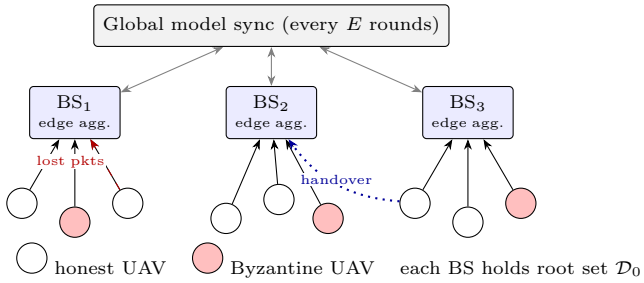


Fig. 1. Schematic of hierarchical UAV-swarm FL: each BS aggregates the (partially received) updates of the UAVs it serves; BS models are synchronized every E rounds.

A natural idea is to let the BS's own link telemetry modulate trust: a UAV on a poor link should not be blamed for an update that looks unusual because of the channel. Our earlier single-seed experiments suggested such a rule beat a cosine-weighted baseline inspired by prior UAV-FL work [11]. This letter tests that idea under controlled conditions, for digital packet erasures in hierarchical UAV-client FL, and finds that it does not hold. The main contributions of this letter are as follows:

1) We model hierarchical UAV-swarm FL whose uplink packet erasures are driven by a measured DeepSense 6G drone received-power trace [12], [13], with BS association, handover and model-poisoning attackers.
2) We propose an erasure-aware reputation aggregator and isolate its two channel-aware mechanisms, erasure masking and telemetry-weighted reputation, in a prespecified, paired 720-run simulation study.
3) We identify the zero-drift equilibrium of the reputation update as an expectile of the behavior score [14] and show that it falls as link evidence weakens, consistent with the observed loss from telemetry.

II. System Model

A. Hierarchical learning

K UAVs train a model $\mathbf{w} \in \mathbb{R}^P$ and are served by B BSs (Fig. 1). In round t, UAV k attached to BS b starts from the edge model $\mathbf{w}_b^t$, runs local SGD on its private data $\mathcal{D}_k$ and uploads $\mathbf{\Delta}_k^t$. BS b aggregates the updates

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Code, summary results and the scripts that produce every figure: https://github.com/sasank-555/uav-swarm-fl. The raw DeepSense 6G data must be obtained from its provider.

it receives into $\mathbf{w}_b^{t+1} = \mathbf{w}_b^t + \mathrm{Agg}(\cdot)$. Every E rounds the BS models are averaged, weighted by each cell's number of training samples, and broadcast [15]. Each BS also holds a small root set $\mathcal{D}_0$, disjoint from all UAV data and assumed clean, on which it computes a reference update $\mathbf{r}_b^t$ each round, as in FLTrust [6].

B. Erasure channel

An update is split into packets of L parameters. The uplink SNR γ_k^t of UAV k follows the relative variation of a measured drone received-power sequence [12], [13], replayed around an assumed reference SNR $\bar{\gamma}$ and adjusted for the 3-D distance to each virtual BS. Conditional on the link state, packets are lost independently, all parameters of a packet sharing one erasure, with probability

$$p_k^t = \big(1 + \exp\big(a(\gamma_k^t - \gamma_{\mathrm{th}})\big)\big)^{-1}, \tag{1}$$

and an update that misses the round deadline is dropped. The BS therefore observes $\tilde{\boldsymbol{\Delta}}_k = \mathbf{m}_k \odot \boldsymbol{\Delta}_k$, where $\mathbf{m}_k \in \{0,1\}^P$ marks delivered coordinates. Association uses a hysteresis-and-time-to-trigger handover rule on received power. There is no jammer and no additive parameter noise.

C. Threat model

A fixed set of f UAVs is Byzantine. They follow the protocol but replace their update by a sign-flip ($-\lambda\boldsymbol{\Delta}_k$), an ALIE-style update [16] ($\boldsymbol{\mu} - z\boldsymbol{\sigma} \odot \mathrm{sign}(\boldsymbol{\mu})$, from the honest updates of the same cell), or an inner-product-manipulation (IPM) update [17] ($-\epsilon\boldsymbol{\mu}$). The ALIE and IPM attackers observe honest updates in their cell, a deliberately strong assumption. Telemetry is measured by the BS and is assumed authentic.

III. Proposed Erasure-Aware Reputation Aggregation

Let $\mathcal{S}(\mathbf{x}; \mathbf{m})$ restrict $\mathbf{x}$ to the coordinates where $\mathbf{m} = 1$. Each BS runs the following steps (index t dropped).

1) Clipping. With ρ_k the RMS of the delivered entries of $\tilde{\boldsymbol{\Delta}}_k$, scale the update by $s_k = \min(1, \mathrm{med}_j\, \rho_j/\rho_k)$ (regularized for $\rho_k = 0$), which limits the effect of large-norm updates.

2) Masked similarity. Compare each update with the root update on delivered coordinates only, which removes the zero-fill artifact of lost packets: $c_k = \cos\big(\mathcal{S}(s_k\tilde{\boldsymbol{\Delta}}_k; \mathbf{m}_k), \mathcal{S}(\mathbf{r}; \mathbf{m}_k)\big)$, with $c_k = 0$ for an empty mask or a zero vector.

3) Behavior score. $b_k = \mathrm{clip}\big(c_k/\mathrm{med}_{j:c_j>0}\, c_j,\, 0,\, 1\big)$, and $b_k = 0$ for all k if no c_j is positive.

4) Reputation. With memory $\beta \in [0, 1)$,

$$T_k \leftarrow T_k + \eta_k\,(b_k - T_k), \qquad \eta_k = \begin{cases} (1-\beta)\, e_k, & b_k > T_k, \\ 1-\beta, & b_k \le T_k, \end{cases} \tag{2}$$

so that bad behavior is penalized at full rate and a weak link cannot shield an attacker that drops its own packets. The evidence $e_k \in [0, 1]$ slows rewards on poor links: $e_k = \mathrm{PDR}_k \cdot R_k \cdot h_k \cdot g_k \cdot v_k$, the product of the packet delivery ratio, long-term reliability, a handover penalty ($h_k = 0.5$ during a handover), a distance factor and a speed factor.

5) Masked weighted average. UAVs with $T_k < \tau$ get zero weight; otherwise $\omega_k = n_k T_k$, with $n_k = |\mathcal{D}_k|$ and a staleness discount. Each coordinate is averaged only over the UAVs that delivered it, and is zero if none did:

$$\mathrm{Agg}_p = \frac{\sum_k \omega_k\, m_{k,p}\, s_k \tilde{\Delta}_{k,p}}{\sum_k \omega_k\, m_{k,p}}. \tag{3}$$

Variants. CCRA (full) uses all steps with telemetry evidence e_k. Mask-only sets $e_k = 1$ but keeps the masks. No-context additionally sets $\mathbf{m}_k = \mathbf{1}$, so lost packets enter steps 1–5 as zeros, which is how a channel-unaware server sees them. Deadline handling is shared by all variants, and the SINR calibration of c_k in our code is inactive here because there is no parameter noise, so the full-versus-mask-only contrast isolates e_k.

IV. Equilibrium Analysis of the Reputation

Consider one UAV with constant evidence $e_k = e \in (0, 1]$ and behavior scores $b \in [0, 1]$ that are i.i.d., independent of its current reputation, with $0 < \mathbb{E}[b] < 1$. By (2), the conditional expected step from level T is $(1-\beta)\, g_e(T)$ with

$$g_e(T) = e\,\mathbb{E}\big[(b-T)^+\big] - \mathbb{E}\big[(T-b)^+\big]. \tag{4}$$

Proposition 1: g_e has a unique zero $T^\star(e) \in (0, 1)$. It equals $\mathbb{E}[b]$ for $e = 1$, is nondecreasing in e, tends to $\operatorname{ess\,inf} b$ as $e \to 0$, and is the expectile of b at level $q = e/(1+e)$ [14].

Proof: g_e is continuous with $g_e(0) = e\,\mathbb{E}[b] > 0$ and $g_e(1) = -\mathbb{E}[1-b] < 0$, so a zero exists. For $T_2 > T_1$, $g_e(T_2) - g_e(T_1)$ lies between $-(T_2 - T_1)$ and $-e(T_2 - T_1)$, so g_e is strictly decreasing and the zero is unique. For $e = 1$, $g_1(T) = \mathbb{E}[b] - T$. Since $g_e(T)$ is nondecreasing in e, a larger e cannot lower the zero. For $T < \operatorname{ess\,inf} b$, $g_e(T) > 0$, so the zero is at least $\operatorname{ess\,inf} b$; for any larger T, $\mathbb{E}[(T-b)^+] > 0$ while $e\,\mathbb{E}[(b-T)^+] \to 0$, so $g_e(T) < 0$ for small e. Dividing $g_e = 0$ by $1 + e$ gives $q\,\mathbb{E}[(b-T)^+] = (1-q)\,\mathbb{E}[(T-b)^+]$, the expectile condition. ∎

$T^\star$ is the zero of the conditional drift, not an almost-sure limit: with $\beta = 0.5$ the reputation keeps fluctuating, and in the simulator e_k and b_k vary over time and depend on other UAVs through the median normalization. Under this approximation, weaker evidence moves the equilibrium from the mean score toward the lower tail of the score distribution. The asymmetry meant to stop packet-dropping attackers can therefore lower the trust of an honest UAV with non-IID data on a weak link, so that it falls below τ more often, while an attacker whose scores are already near zero is suppressed little further. Section V checks whether the gate statistics are consistent with this mechanism.

V. Simulation Results

A. Simulation Setup

Table I lists the values assumed for the parameters. Attackers are chosen per seed and kept fixed across

TABLE I
Values Assumed for the Parameters

Parameter	Value
UAVs K; BSs B; attackers f; sync E	10; 3; 3; 5
Rounds; local training	100; one SGD epoch
Model	CNN, 156k parameters
Data	aerial-transformed CIFAR-10, Dirichlet(0.5) [18]
Root set $\lvert\mathcal{D}_0\rvert$; β; τ	500 images; 0.5; 0.3
Packet size L; a; γ_{th}	1024; 1; 3 dB
Reference SNR $\bar{\gamma}$	ideal, 15 dB, 5 dB
Sign flip λ; ALIE z; IPM ϵ	5; 1.5; 0.5
Seeds; runs	10; 720

TABLE II
Run outcomes over all 720 runs (120 per method)

Method	Diverged	All runs (%)	Attacked, 5 dB (%)
FedAvg	41/120	37.3	29.4
Median	26/120	29.3	21.3
FLTrust	0/120	50.6	50.1
No-context	0/120	51.4	45.3
Mask-only	0/120	52.8	49.5
CCRA (full)	0/120	51.1	46.0

methods. The channel is either ideal or the same measured trace replayed at assumed anchors $\bar{\gamma} = 15$ dB (mean PDR 99.8%) and $\bar{\gamma} = 5$ dB (mean PDR 65.0%). The baselines are FedAvg [1], coordinate-wise median [5] and FLTrust [6], which we adapt to the masks: the median ignores erased entries, and FLTrust scores cosine similarity and matches the root norm on delivered coordinates, then averages coordinate-wise. FLTrust shares the root set and a matched local-step budget with CCRA.

The full factorial design crosses six aggregators, four attack settings (including clean), three channels and 10 paired seeds ($6 \times 4 \times 3 \times 10 = 720$). A run that diverges numerically is scored at chance (10%), never discarded. The primary contrast, CCRA minus mask-only at 5 dB minus the same at 15 dB, and its 1 pp decision threshold were fixed before the runs. Intervals are 95% percentile bootstraps over seeds (10,000 resamples). All other contrasts are exploratory and not adjusted for multiple comparisons.

A separate exploratory 240-run follow-up tests packet-level fresh-only and reuse-old rules [3] on matched seeds 41–50. Fresh-only averages delivered coordinates; reuse-old fills erasures from each client's last received parameters. Both retain the unused 500-image root reserve.

B. Robustness and Stability

Fig. 2 and Table II show that FedAvg and the coordinate-wise median diverged in 41 and 26 of their 120 runs respectively (FedAvg in all 30 sign-flip runs), whereas none of the four root-referenced methods diverged in its 120 runs. Across all runs, mask-only attains the highest mean accuracy (52.8% versus 37.3% for FedAvg). Against FLTrust it gains 2.6 pp ([0.9, 4.5]) under attack at

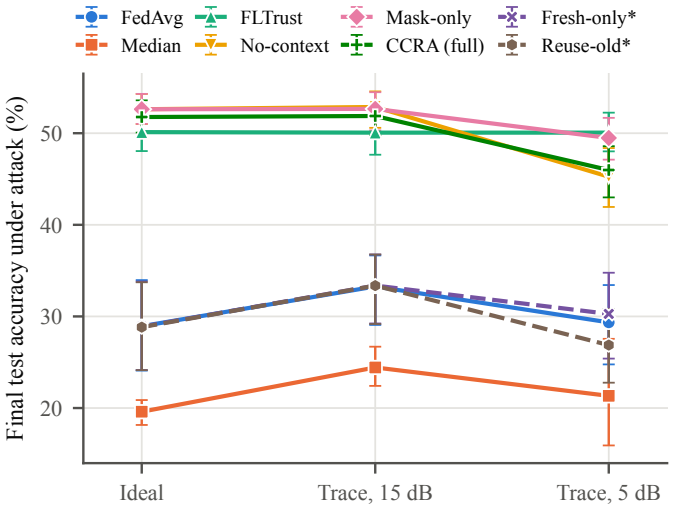


Fig. 2. Final accuracy under attack (mean of sign-flip, ALIE and IPM over 10 seeds; error bars: 95% seed-bootstrap intervals). Dashed, starred rules come from the separate 240-run follow-up. Diverged runs count as 10%.

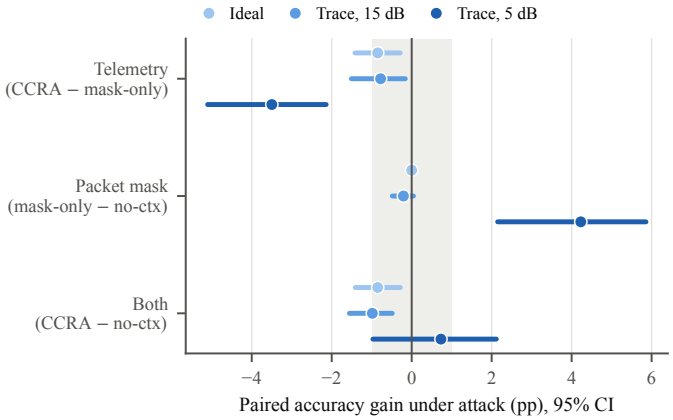


Fig. 3. Paired ablation contrasts under attack. The gray band marks the prespecified ±1 pp region of practical equivalence.

15 dB, while the 5 dB comparison is inconclusive (−0.6 pp, [−2.0, 1.0]; exploratory).

C. Which Channel Awareness Helps

Fig. 3 separates the two mechanisms. The prespecified stress interaction is −2.7 pp ([−3.9, −1.5]), below the −1 pp practical threshold. On the ideal channel the masks are all ones, so mask-only and no-context coincide. As delivery degrades, erasure masking becomes valuable: at 5 dB it adds 4.2 pp ([2.2, 5.9]). Telemetry does the opposite: at 5 dB, full CCRA minus mask-only is −3.5 pp ([−5.1, −2.1]; 46.0% versus 49.5%). Combining the two mechanisms leaves an inconclusive net effect relative to no-context.

D. Consistency with the Equilibrium Analysis

Fig. 4 is consistent with Proposition 1. With telemetry, the gate rejects 41.2% of honest updates versus 13.8%

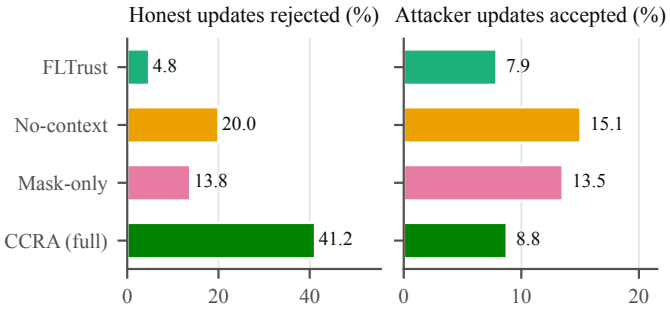


Fig. 4. Reputation-gate decisions under attack at 5 dB, averaged over attacks and seeds (percent of received updates).

for mask-only, and admits only slightly fewer attacker updates (8.8% versus 13.5%). The rule trades a small gain in attacker filtering for a large loss of honest data where honest data is already scarce. FLTrust rejects only 4.8% of honest updates; it has no reputation memory, but its norm matching, score normalization and weights also differ, so this does not isolate the role of memory. These rates are descriptive and do not by themselves establish the causal mechanism.

E. Base-Paper Erasure Rules

Fresh-only and reuse-old diverged in 41/120 and 42/120 runs, respectively, including all 30 sign-flip runs each. On clean 5 dB links they attain 56.86% and 51.47%, a paired fresh-only gain of 5.39 pp ([4.11, 6.74]). Under attack at 5 dB, the gain is 3.40 pp ([0.94, 6.60]). These exploratory intervals retain divergent seeds at chance. The rules address erasures, not Byzantine clients; this packet-level, hierarchical experiment does not test the original paper's convergence theorem.

F. Limitations

We use one measured BS with virtual neighbors, assumed SNR anchors, few handovers, a clean root set, one model and aerial-transformed CIFAR-10. Telemetry spoofing and adaptive reputation attacks [10] are untested. The follow-up used GTX 1650/RTX 2060 workers, differing software and checkpoint recovery, versus an A100 ablation; cross-study rankings are descriptive, without hardware-controlled inference.

VI. Conclusion

With measured-trace packet erasures, masking improved attacked accuracy, while telemetry-weighted reputation hurt weak-link performance and rejected more honest updates. Fresh-only also outperformed reuse-old on clean stressed links. Future work will redesign the reward rule and test fresh seeds, stronger mobility and adaptive attackers.

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